

# Linear Methods for Classification

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```
rm(list = ls())
library(readr) ; library(tidyverse) ; library(corrplot)

train <- read.csv("vowel_train.csv", sep = ",")
train$x.10. <- gsub("\\\\", "", train$x.10.)
train$x.10. <- as.numeric(gsub("}", "", train$x.10.))
train$x.10 <- train$x.10. ; train$x.10. <- NULL

test <- read.csv("vowel_test.csv", sep = ",")
test$x.10. <- gsub("\\\\", "", test$x.10.)
test$x.10. <- as.numeric(gsub("}", "", test$x.10.))
test$x.10 <- test$x.10. ; test$x.10. <- NULL
head(train) ; summary(train)
```

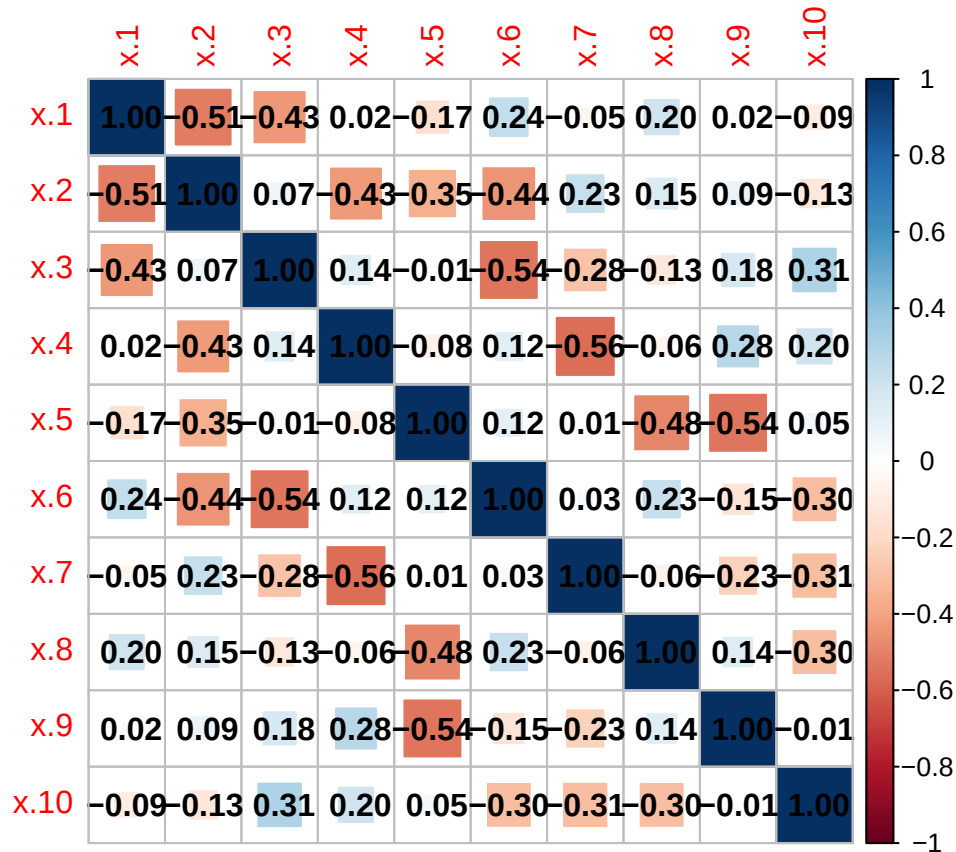
```
##   y    x.1    x.2    x.3    x.4    x.5    x.6    x.7    x.8    x.9    x.10
## 1 1 -3.639 0.418 -0.670 1.779 -0.168 1.627 -0.388 0.529 -0.874 -0.814
## 2 2 -3.327 0.496 -0.694 1.365 -0.265 1.933 -0.363 0.510 -0.621 -0.488
## 3 3 -2.120 0.894 -1.576 0.147 -0.707 1.559 -0.579 0.676 -0.809 -0.049
## 4 4 -2.287 1.809 -1.498 1.012 -1.053 1.060 -0.567 0.235 -0.091 -0.795
## 5 5 -2.598 1.938 -0.846 1.062 -1.633 0.764 0.394 -0.150 0.277 -0.396
## 6 6 -2.852 1.914 -0.755 0.825 -1.588 0.855 0.217 -0.246 0.238 -0.365

##           y           x.1           x.2           x.3
## Min.      : 1   Min.      : -5.211   Min.      : -1.2740   Min.      : -2.48700
## 1st Qu.: 3     1st Qu.: -3.923   1st Qu.: 0.9167   1st Qu.: -0.94550
## Median : 6     Median : -3.097   Median : 1.7330   Median : -0.50250
## Mean      : 6     Mean      : -3.167   Mean      : 1.7353   Mean      : -0.44800
## 3rd Qu.: 9     3rd Qu.: -2.512   3rd Qu.: 2.4038   3rd Qu.: 0.04925
## Max.      :11    Max.      : -0.941   Max.      : 5.0740   Max.      : 1.41300

##           x.4           x.5           x.6           x.7
## Min.      : -1.4090   Min.      : -2.1270   Min.      : -0.8360   Min.      : -1.53700
## 1st Qu.: -0.0835   1st Qu.: -0.9307   1st Qu.: 0.1085   1st Qu.: -0.29700
## Median : 0.4565   Median : -0.4170   Median : 0.5275   Median : 0.04000
## Mean      : 0.5250   Mean      : -0.3893   Mean      : 0.5850   Mean      : 0.01748
## 3rd Qu.: 1.1640   3rd Qu.: 0.1155   3rd Qu.: 1.0097   3rd Qu.: 0.34800
## Max.      : 2.1910   Max.      : 1.8310   Max.      : 2.3270   Max.      : 1.40300

##           x.8           x.9           x.10
## Min.      : -1.29300   Min.      : -1.6130   Min.      : -1.68000
## 1st Qu.: -0.01825   1st Qu.: -0.6737   1st Qu.: -0.50700
## Median : 0.47700   Median : -0.2550   Median : -0.08250
## Mean      : 0.41739   Mean      : -0.2681   Mean      : -0.08457
## 3rd Qu.: 0.86125   3rd Qu.: 0.1375   3rd Qu.: 0.30100
## Max.      : 1.67300   Max.      : 1.3090   Max.      : 1.39600
```

```
corr_matrix <- cor(train[,-1]) # correlation matrix
corrplot(corr_matrix, method = "square", addCoef.col = 'black') # associated plot
```



```
train[paste0("y_", 1:11)] <- lapply(1:11, function(k) { as.integer(train$y == k)})
test[paste0("y_", 1:11)] <- lapply(1:11, function(k) { as.integer(test$y == k)})

x_vars <- paste0("x.", 1:10) # string vector of the name of the features

models <- vector("list", 11) # create an empty list to store the 11 regressions

for (k in 1:11) { # loop through the 11 groups
  y_var <- paste0("y_", k) # string vector of the name of group k

  formula_k <- as.formula(
    paste(y_var, "~", paste(x_vars, collapse = " + ")) # formula regress y_k on the features
  )

  models[[k]] <- lm(formula_k, data = train) # fit the reg
}

fitted_train <- as.data.frame(
  lapply(models, fitted) # retrieve the fitted values for each model
) # we store them in a data.frame

names(fitted_train) <- paste0("y_", 1:11) # rename the columns for clarity
```

```

# the predicted outcome is the one with the highest fitted values
prediction_train <- max.col(fitted_train, ties.method = "first")

# Training error rate is ~0.48
train_error_rate <- mean(train$y != prediction_train) ; train_error_rate

## [1] 0.4772727

# we obtain the fitted values of the model on the test data
fitted_test <- as.data.frame(
  lapply(models, function(model) predict(model, newdata = test))
)

names(fitted_test) <- paste0("y_", 1:11)

prediction_test <- max.col(fitted_test, ties.method = "first")

# the test error rate is ~0.67
test_error_rate <- mean(test$y != prediction_test) ; test_error_rate

## [1] 0.6666667

```